Prompt Engineering for Software Engineering: Summary of PROMPT-SE 2026

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Abstract

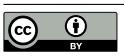
Prompt engineering has become a fundamental component of language-model-based software engineering, influencing the quality, reliability, reproducibility, and cost of AI-assisted development activities. As research in this area rapidly expands, empirical studies remain fragmented across software engineering tasks, evaluation methodologies, and application domains, highlighting the need for a dedicated forum to consolidate emerging knowledge and research practices. The First International Workshop on Empirical Prompt Engineering for Software Engineering (PROMPT-SE), co-located with EASE 2026, was established to address this need by bringing together researchers and practitioners working on the empirical study of prompt and instruction artifacts. The workshop featured contributions spanning multiple software engineering activities, including requirements engineering, software development, software repository analysis, and qualitative research, while also addressing broader topics such as empirical evaluation, multi-agent systems, governance, and sustainability. This paper summarizes the workshop, synthesizes the main research themes that emerged across the accepted contributions, and discusses their implications for the evolution of empirical prompt engineering as an emerging area of software engineering research.

CCS Concepts

• **Software and its engineering** → **Software creation and management**; **Empirical software validation**; • **Computing methodologies** → *Natural language processing*; • **General and reference** → *Empirical studies.*

Keywords

Empirical Software Engineering, Prompt Engineering, Large Language Models, Software Engineering.



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1 Introduction

Language Models (LMs), including both Large Language Models (LLMs) and Small Language Models (SLMs), are reshaping Software Engineering (SE), supporting activities such as requirements engineering, software development, testing, maintenance, and empirical research [2, 4]. Beyond assisting developers in individual tasks, recent advances in agentic AI have enabled LM-based systems to coordinate multiple agents, tools, APIs, and external knowledge sources, further expanding their role throughout the software development lifecycle [1, 3].

As these systems become more sophisticated, their behavior depends on the quality of the artifacts used to specify and guide model interactions. These artifacts—including prompts, system prompts, contextual information, message templates, and interaction protocols—influence the effectiveness, reliability, reproducibility, and cost of LM-based SE processes. As a result, they are being regarded as reusable engineering artifacts whose design, evaluation, and maintenance deserve systematic investigation [5].

Despite the rapid growth of research on prompt engineering, empirical evidence remains fragmented across application domains, SE tasks, evaluation methodologies, and model families. While numerous studies have investigated prompting techniques for specific problems, there are fewer opportunities for researchers to discuss common empirical practices, shared evaluation methodologies, and broader research challenges across the field [6, 7]. This fragmentation makes it difficult to compare results across studies, identify reusable engineering practices, and establish a cumulative body of knowledge for prompt engineering in SE.

To address this need, the *First International Workshop on Empirical Prompt Engineering for Software Engineering (PROMPT-SE*

2026) [1], co-located with the International Conference on Evaluation and Assessment in Software Engineering (EASE 2026) [2], was established as a dedicated forum for researchers and practitioners interested in the empirical study of prompt engineering and related instruction artifacts in SE. Consistent with the empirical focus of EASE, the workshop combined technical presentations, interactive discussions, and community-building activities to encourage the exchange of ideas, identify emerging research themes, and foster collaborations across different SE domains.

This extended abstract summarizes the first edition of PROMPT-SE by presenting an overview of the workshop, synthesizing the main research themes from the accepted contributions, and discussing their implications for the continued evolution of empirical prompt engineering as a research area within SE.

2 Workshop Overview

PROMPT-SE 2026 was organized as a one-day workshop co-located with the 30th International Conference on Evaluation and Assessment in Software Engineering (EASE 2026). The workshop was designed to provide a dedicated venue for discussing empirical research on prompt engineering and related instruction artifacts, with a particular emphasis on evaluation methodologies, evidence-based practices, and the role of LMs in modern SE.

The workshop received 11 submissions, one of which was withdrawn before the review process. Submissions were evaluated according to the EASE review criteria, namely soundness, significance, novelty, verifiability, transparency, and presentation quality. These criteria allowed the Program Committee to assess both the scientific contribution of each paper and the extent to which its methods, data, analysis, and presentation supported rigorous empirical research. The papers underwent a rigorous peer-review process involving at least three reviewers per submission. After this process, six papers were accepted for presentation, comprising four full papers and two short papers.

The technical program combined paper presentations with interactive question-and-answer sessions, encouraging discussion on both technical contributions and broader methodological challenges. The accepted papers covered a diverse range of SE activities, including requirements engineering, software development, software repository analysis, qualitative research, and software sustainability. At the same time, several contributions addressed cross-cutting concerns such as prompt evaluation, multi-agent systems, governance, and empirical assessment methodologies, illustrating the breadth of current research on prompt engineering for SE.

The workshop concluded with a structured breakout session designed to encourage participants to collectively reflect on current practices, methodological challenges, and future research opportunities. Rather than focusing solely on individual research contributions, the activity encouraged participants to reflect on common challenges and to exchange perspectives across different SE domains.

Overall, PROMPT-SE 2026 attracted approximately 15–20 participants and provided an interactive environment that combined technical presentations with community discussion, contributing to the consolidation of an emerging research community around empirical prompt engineering for SE.

[1] https://conf.researchr.org/home/ease-2026/prompt-se-2026
[2] https://conf.researchr.org/home/ease-2026

3 Emerging Research Themes

The accepted papers covered a broad spectrum of SE activities, ranging from requirements engineering and software development to software repository analysis, qualitative research, and software sustainability. Although these contributions addressed different research questions and application domains, they collectively revealed several recurring themes that characterize the current evolution of empirical prompt engineering. Rather than presenting isolated studies on the use of LMs, the accepted contributions highlighted common methodological concerns, emerging interaction paradigms, and shared research challenges that cut across traditional SE boundaries.

3.1 Prompt Engineering Across the Software Engineering Lifecycle

One of the clearest observations from the workshop was the breadth of SE activities in which prompt engineering is now being investigated. The accepted contributions demonstrated that LMs are increasingly supporting not only software implementation, but also requirements engineering, qualitative data analysis, software repository mining, and developer assistance. This diversity suggests that prompt engineering is rapidly evolving into a transversal concern spanning the entire SE lifecycle, extending well beyond traditional code-centric tasks.

Despite addressing different application domains, the accepted papers shared a common objective: understanding how prompting strategies affect the effectiveness of LMs under specific SE conditions. Rather than simply proposing new prompting techniques, most contributions investigated *when*, *why*, and *under which circumstances* different strategies provide measurable benefits. This empirical perspective shifts the focus from ad hoc prompt optimization toward systematic evaluation and evidence-based engineering practices.

3.2 Increasing Complexity of Prompt-Based Interactions

A second recurring theme concerned the growing complexity of interactions between developers and LMs. Several contributions moved beyond individual prompts by investigating structured workflows involving iterative refinement, context management, external knowledge sources, and cooperation among multiple LMs or agents. These studies illustrate that effective LM-based SE increasingly depends on orchestrating multiple interaction steps rather than optimizing a single prompt in isolation.

This increasing complexity also raises human-centered questions. Software engineers must understand, inspect, adapt, and trust prompt-based workflows, making usability, explainability, and developer control important factors in the design of LM-based SE tools.

More broadly, this trend reflects an evolution in how prompts are perceived within SE processes. Instead of being treated as standalone textual inputs, prompts are becoming integral components

of larger interaction workflows that combine contextual information, tool invocation, collaborative reasoning, and iterative feedback. As these interaction paradigms become more sophisticated, understanding how to systematically design, evaluate, and maintain them represents an important research direction.

Several discussions suggested that the field is gradually moving from prompt engineering—understood as the design of individual textual inputs—toward instruction engineering, where prompts, templates, system instructions, tool-use policies, and agent communication protocols are treated as interconnected artifacts within LM-based software systems.

3.3 Empirical Evaluation as a Common Ground

Although the accepted papers investigated different SE tasks, they shared a common methodological perspective centered on empirical evaluation. Controlled experiments, comparative analyses, benchmark datasets, human evaluation, case studies, and mixed empirical methods were consistently employed to assess the effectiveness of prompting strategies. This methodological diversity reflects the growing recognition that prompt engineering should be studied through rigorous empirical investigation rather than anecdotal evidence or isolated demonstrations of LM capabilities.

A recurring point in the discussions was that evaluating prompt-based techniques introduces specific methodological challenges that go beyond traditional empirical evaluation. The same prompting strategy may lead to different outcomes depending on the model version, execution parameters, task formulation, input context, and evaluation setting. For this reason, participants emphasized the importance of clearly reporting prompts, model configurations, datasets, baselines, evaluation metrics, and the number of runs performed. Such reporting practices are essential to make empirical results interpretable, comparable, and reusable by the community.

This perspective closely aligns with the goals of the EASE community, emphasizing evidence-based SE research and systematic evaluation of emerging AI technologies. More importantly, the workshop discussions highlighted that meaningful progress depends not only on improving prompting strategies but also on developing reproducible experimental protocols, transparent reporting practices, and evaluation methodologies that enable cumulative scientific knowledge.

3.4 Emerging Research Challenges

Beyond the individual technical contributions, the workshop revealed several research challenges that are common across application domains. Regardless of the specific SE task being addressed, many studies encountered similar issues related to evaluation, reproducibility, transferability of results across models and datasets, and the difficulty of comparing alternative prompting strategies under consistent experimental settings. These observations reinforce the need for shared evaluation protocols, benchmark suites, and reporting guidelines to support more rigorous and comparable empirical studies.

Another challenge concerns the engineering lifecycle of prompt-based solutions. As prompts become increasingly embedded within complex LM-based systems, research attention is gradually shifting from designing effective prompts to understanding how prompt and instruction artifacts can be validated, documented, versioned, maintained, and integrated into SE workflows. This observation suggests that prompts and related instruction artifacts should be managed according to principles similar to those applied to other software artifacts (e.g., source code, requirements specifications, or test cases), particularly when they are part of development workflows or agentic systems.

Collectively, the accepted contributions suggest that empirical prompt engineering is moving beyond isolated application studies toward a broader research agenda focused on establishing engineering principles, empirical methodologies, and best practices for the development of reliable LM-based software systems.

4 Reflections and Future Directions

The first edition of PROMPT-SE demonstrated the value of establishing a dedicated forum for discussing empirical research on prompt engineering in SE. While the accepted papers addressed different application domains and research questions, the workshop revealed a strong convergence around common methodological concerns, highlighting the need for shared evaluation practices and a stronger dialogue across traditionally separated research communities.

A key outcome of the workshop was the recognition that many challenges associated with LM adoption extend beyond individual SE tasks. Issues such as prompt evaluation, reproducibility, experimental design, context management, and integration into development workflows emerged as recurring concerns across multiple contributions. This suggests that advancing prompt engineering requires not only task-specific innovations but also a broader engineering perspective capable of supporting the systematic development and empirical assessment of LM-based solutions.

The workshop also confirmed the importance of fostering collaboration between researchers and practitioners working on different aspects of LM adoption in SE. By bringing together expertise from empirical SE, artificial intelligence, requirements engineering, software analytics, and human-centered SE, PROMPT-SE encouraged discussions that would be difficult to achieve within more specialized venues. The interactive format of the workshop, including paper presentations and collaborative discussions, further strengthened the exchange of perspectives and helped identify common research interests spanning application domains.

Looking ahead, the continued evolution of LMs, agentic systems, and complex interaction paradigms will introduce new challenges for empirical SE research. Future work will need to address not only the effectiveness of prompting strategies but also their reliability, maintainability, reproducibility, and integration into SE processes. Based on the workshop contributions and discussions, we identify four priorities for future research: (i) developing reporting guidelines for prompt-based studies, (ii) designing benchmarks that capture realistic SE tasks, (iii) studying the evolution and maintenance of prompts in software projects, and (iv) evaluating prompt-based systems beyond accuracy, including cost, reliability, security, energy consumption, and human oversight.

The first edition of PROMPT-SE suggests that empirical prompt engineering is evolving beyond isolated application studies toward a more cohesive research area grounded in SE principles. In this context, the workshop provides an initial foundation for consolidating

a research community around evidence-based prompt engineering in SE.

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